

TASK 1: Download Files from Amazon S3 to an EC2 Instance

Objective

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The objective of this lab is to learn how to:

- Launch an EC2 instance
- Configure access permissions for S3
- Install AWS CLI on EC2
- Download files from an S3 bucket to the EC2 instance

Architecture Overview

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Amazon S3 Bucket → EC2 Instance → Local Directory

The EC2 instance will fetch files stored in the S3 bucket using AWS CLI commands.

Prerequisites

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Students must have:

- AWS Account
- Basic knowledge of Linux commands
- AWS services
- AWS IAM permissions to create EC2, IAM Role, and S3 Bucket

Task Steps

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Step 1: Create an S3 Bucket

1. Login to AWS Console
2. Go to S3 Service

3. Click Create Bucket

4. Enter:

Bucket name: student-s3-download-lab

Region: same as EC2 region

5. Keep default settings

6. Click Create Bucket

Step 2: Upload Files to S3

1. Open the bucket student-s3-download-lab

2. Click Upload

3. Upload any test files such as:

sample1.txt

sample2.txt

data.csv

Step 3: Create an IAM Role for EC2

1. Go to IAM Service

2. Click Roles

3. Click Create Role

4. Select:

Trusted entity: AWS Service

Use case: EC2

5. Attach policy:

AmazonS3ReadOnlyAccess

6. Role name:

EC2-S3-ReadRole

7. Click Create Role

Step 4: Launch an EC2 Instance

1. Go to EC2 Dashboard
2. Click Launch Instance

Configuration:

Name: S3-Download-Lab

AMI: Ubuntu 22.04 or Amazon Linux

Instance Type: t2.micro

Key Pair: Create or select existing

3. Under Advanced Settings attach IAM Role:

EC2-S3-ReadRole

4. Launch the instance

Step 5: Connect to EC2

Use SSH:

```
ssh -i mykey.pem ubuntu@<EC2-PUBLIC-IP>
```

For Amazon Linux:

```
ssh -i mykey.pem ec2-user@<EC2-PUBLIC-IP>
```

Step 6: Install AWS CLI on EC2

Ubuntu:

```
sudo apt update
```

```
sudo apt install awscli -y
```

Amazon Linux:

```
sudo yum install awscli -y
```

Verify installation:

```
aws --version
```

Step 7: List Files in S3 Bucket

```
aws s3 ls s3://student-s3-download-lab
```

Expected output example:

```
sample1.txt
```

```
sample2.txt
```

```
data.csv
```

Step 8: Download a Single File from S3

```
aws s3 cp s3://student-s3-download-lab/sample1.txt .
```

Check the file:

```
ls
```

Step 9: Download Entire Folder from S3**Create directory:**

```
mkdir downloads
```

Download all files:

```
aws s3 sync s3://student-s3-download-lab downloads/
```

Verify:

```
ls downloads
```

Step 10: Bash Script to Automate Download**Create script:**

```
nano download_s3.sh
```

Script content:

```
#!/bin/bash  
  
S3_BUCKET="s3://student-s3-download-lab"  
  
LOCAL_DIR="/home/ubuntu/s3files"  
  
echo "Creating directory..."  
  
mkdir -p $LOCAL_DIR  
  
echo "Downloading files from S3..."  
  
aws s3 sync $S3_BUCKET $LOCAL_DIR  
  
echo "Download completed"
```

Make executable:

```
chmod +x download_s3.sh
```

Run script:

```
./download_s3.sh
```

Expected Output

Files should be downloaded to:

```
/home/ubuntu/s3files
```

Verify:

```
ls /home/ubuntu/s3files
```

Bonus Tasks

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1. Download only specific file types:

```
aws s3 sync s3://student-s3-download-lab downloads/ --exclude "*" --include "*.txt"
```

2. Schedule automatic download using cron:

`crontab -e`

Example:

```
0 * * * * /home/ubuntu/download_s3.sh
```

Learning Outcomes

After completing this lab, students will understand:

- How EC2 accesses S3 using IAM roles
- How to use AWS CLI for S3 operations
- How to automate S3 downloads using Bash scripts